

Development and Validation of AI Based Model for Myopia Prediction Compared with Conventional Clinical Refraction



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Synopsis

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Title of Research: Development and Validation of AI Based Model for Myopia Prediction
Compared to with Conventional Clinical Refraction

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Development and Validation of AI Based Model for Myopia Prediction Compare with Conventional Clinical Refraction

Abstract

Myopia will continue to be a major global public health issue, especially in young adults and children. Serious visual consequences such as retinal detachment, glaucoma, and myopic maculopathy are linked to progressive myopia, underscoring the importance of early detection and prompt treatment. In optometric practice, corneal topography will be often utilized to evaluate corneal curvature and surface features; however, traditional clinical interpretation of topographic maps will continue to be subjective and reliant on practitioner experience, potentially limiting predictive reliability. Ophthalmic imaging data analysis will become more objective and data-driven thanks to recent developments in artificial intelligence (AI). AI based corneal topography analysis will make it possible to automatically identify complicated patterns and tiny corneal changes that could be difficult to spot with standard clinical examination. In order to forecast the evolution of myopia, this study will compare AI based corneal topography analysis with traditional clinical examination. Corneal topography and refractive data from myopic patients will be used in a comparative analytical study design. Accuracy, sensitivity, specificity, and area under the receiver operating characteristic curve will be used to assess the predictive performance of both methods. After the completion of this study the results are expected to determine if AI based corneal topography analysis will function as a useful tool in clinical optometry and provide prognostic validity.

Need for the Research

Owing to myopia promptly increasing prevalence, especially in children and young adults, myopia one of the most common refractive problem in the world has turn up as a notable public health distress. In accord with the epidemiological data, the prevalence of myopia and high myopia is increasing terribly globally, which could have damaging long-term effect on eye health (Liu et al., 2025). Extreme axial elongation of the eye is connected to advance myopia, which intoxicate the risk of visual disorders such cataract, glaucoma, retinal detachment, and myopic maculopathy (Tang et al., 2020). Primitive findings and spectrum prediction of myopia are important for decreasing future visual problems.

Changes in spherical equivalent refraction and axial length measurements are the key indications of myopia development in clinical optometric practice. Corneal topography offers important information about corneal curvature, surface regularity, and biomechanical activity that may contribute to refractive changes, despite being historically employed for contact lens fitting and corneal assessment (Xu et al., 2024). During normal clinical trials, noted by the practitioner's and visual assessment of colored imaging and numerical data play a vital role in the illustration of corneal topographic images. This tendency may going towards inter examiner inconstancy and restrict the ability to detect hypocritical corneal changes related with early myopia progression (Xu et al., 2024). Advances in artificial intelligence (AI), particularly machine learning and deep learning techniques, have opened up new possibilities for objective and data-driven analysis of ocular imaging. Machine learning algorithms can interpret large volumes of high dimensional information and identify complex figures that are usually subtle to the human eye (Zhang et al., 2025). Machine learning has been easily applicable in ophthalmology and optometry in a number of areas now a days, sickness screening, diagnosis, and progression prediction include in this advance area (Liu et al., 2025).

AI based corneal topography analysis has attested positive advancements in diagnostic precision. Deep learning data sets, particularly convolutional neural networks (CNN), have given advanced outputs in the analysis of corneal curvatural imaging data against to ordinary analytical methods (Xu et al., 2024). Various studies have operating AI models to predict axial length enlargement and treatment plans in myopic patients, especially in the evidence of orthokeratology based myopia control (Xu et al., 2025).

Instead of comparing clinical validation, the majority of studies that are currently available concentrate on model construction. Therefore, the goal of this study is to evaluate the relative effectiveness of conventional clinical analysis and AI-based corneal topography analysis in predicting the course of myopia and to explore the potential application of AI as a clinical decision support tool in optometric practice.

Rationale

Due to the increased prevalence of myopia, urgent myopia management measures must be led by precise and early progression prediction. In clinical settings, the interpretation of corneal topography is subjective and may vary depending on the practitioner's experience. AI-based techniques offer a reliable and unbiased alternative that could enhance clinical judgment (Zhang et al., 2025). In contrast to conventional clinical analysis, this study will demonstrate if AI based corneal topography analysis delivers superior prediction accuracy in optometric practice.

Problem Statement

Predicting the progression of myopia using conventional clinical study of corneal topography is subjective and susceptible to observer variability. The ability of AI-based corneal topography analysis and clinical analysis to forecast the progression of myopia has not been thoroughly studied.

Objectives

- To develop artificial intelligence model, including machine learning algorithms, capable for analyzing corneal topography data for the detection early myopia progression.
- To compare the performance of developed models with conventional manual diagnostic.

Hypotheses

Null hypothesis (H_0): AI based corneal topography analysis and clinical analysis do not significantly vary in their ability to predict the progression of myopia.

Alternative hypothesis (H_1): AI based corneal topography analysis outperforms clinical analysis in predicting the progression of myopia.

Operational definitions:

Myopia: Myopia is classified as a spherical equivalent (SE) refraction of at least -0.50DS (mild: -0.50DS to -2.75DS, moderate: -3.00DS to -5.75DS, high: -6.00DS and above).

Corneal Topography: In order to evaluate corneal form and optical quality, corneal topography is a computerized diagnostic technique that quantitatively maps the curvature and shape of the anterior corneal surface. It provides numerical parameters such as central and simulated keratometry values (K1, K2), average corneal power, corneal astigmatism, surface asymmetry index (SAI), surface regularity index (SRI), corneal eccentricity (e-value), and elevation data.

Review of Literature

With a fast rising frequency among children and young adults, myopia has become a significant global visual health concern. Continuous axial elongation of the eyeball is a hallmark of progressive myopia, which increases refractive error and increases the risk of pathological ocular problems in later life (Liu et al., 2025). Myopia progression prognosis is important for early treatment and effective myopia management plan. Corneal topography is crucial for optometric evaluation because it offers a wealth of information about corneal curvature, shape, and surface regularity. The main causes of development of refractive error i.e., myopia are axial length but changes in corneal shapes are also involving in it (Xu et al., 2024). In conventional clinical interpretation of corneal topographic maps, the visual assessment of color-coded patterns and numerical indices is inherently subjective and dependent on practitioner experience. The ordinary clinical trials are very limited and not advances in detecting early and subtle changes in relevance to myopia development (Xu et al., 2024).

The interpretation of ocular imaging data has evolved as a result of recent breakthroughs in artificial intelligence (AI). Machine learning models process large and complex datasets to identify patterns that are difficult for human observers to recognize (Zhang et al., 2025). Machine learning models offers advance methods including objectivity, consistency, and reduced observer bias, making them promising tools for clinical trial support in optometry and ophthalmology (Liu et al., 2025).

The advancement of machine learning corneal topography analysis has been shown in number of studies. Whenever there is a comparison between the conventional methods with machine learning the accuracy increased in machine learning data analysis (Xu et al., 2024). According to Pan et al.,

AI driven frameworks might correctly evaluate corneal therapy zones after orthokeratology, exhibiting AI's power to precisely examine minute corneal modifications (Tang et al., 2021). Machine learning methods has been widely used in forecasting the myopia progression and treatment, whenever the data of axial length in myopic children given, machine learning model gives very accurate results as analyzed through clinical trials (Tang et al., 2020). Recent studies have incorporated corneal topography data into deep learning models to predict myopia control outcomes in orthokeratology patients, with encouraging results (Xu et al., 2025). To boost forecast accuracy and consistency, these algorithms make use of high-dimensional corneal data.

Despite these promising findings, there are still few direct comparisons between AI-based corneal topography analysis and traditional clinical interpretation for myopia progression prediction. Most current research focuses on model creation or treatment outcome prediction rather than comparing AI to conventional clinical analysis (Zhang et al., 2025).

The use of AI in retinal and optic nerve imaging has been shown, but its application in managing refractive errors and predicting the progression of myopia is growing quickly. These advancements will probably influence optometry's future clinical practice. Using population-based clinical data, machine learning (ML) and deep learning (DL) algorithms have been used to forecast future refractive results. With coefficients of determination and (AUC) values consistently above acceptable ranges for clinical application, for example, a large multivariate predictive model created using extensive cohorts spanning infants to young adults produced extremely accurate forecasts of myopia progression and risk of high myopia across validation datasets. This implies that customized clinical approaches for managing myopia will be greatly aided by data-driven models (Li et al., 2024).

Additionally, precision myopia care will be supported more and more by interpretable AI models. A recent study showed how a decision tree technique may be used to stratify children receiving atropine therapy by identifying decision rules based on treatment factors and baseline refractive error that might guide customized follow-up and therapy intensification plans. It is expected that these explainable AI frameworks would promote evidence-based management decisions and increase physician trust (Chen et al., 2025).

Methodology

Study design

This study will be structured as a cross sectional analytical for AI model development and performance evaluation against standard clinical refraction.

Study Place

The study will be carried out in a tertiary eye care facility MTH and optometry clinic Lahore Eye Care Faisalabad that regularly maintains corneal topography data and electronic medical records by using Galilei G6.

Duration of the study

The duration of the study will be from January 2026 to May 2026.

Study population

- Patients with myopia who have had repeated corneal topography exams during a predetermined follow up period will make up the study population.
- Since this age range corresponds to the most active stage of myopia progression, records of patients between the ages of 15 and 25 will be included. Patients of both sexes will be taken into consideration.
- Patients will be eligible for inclusion if they have complete baseline and follow up data and a minimum follow up period of 6 to 7 months.

Sample size

- The clinical database's suitable patient records will be chosen using a non-probability purposive sampling technique.
- The availability of complete longitudinal data and previous research of a comparable nature will be used to decide the sample size, with the goal of include a large enough dataset to guarantee accurate AI model training and statistical comparison.
- If a patient meets the inclusion requirements, both eyes may be included; however, inter eye correlation will be taken into consideration through statistical adjustments. It will be 100 to 150 patients.

Data collection

- Patient records will be used to obtain clinical and imaging data, K values, e-values, Q-values, age and gender, baseline refractive error determined by refraction, axial length measurements when available, and serial corneal topography data including central corneal thickness, anterior chamber depth, lens vault, angle-opening distance, trabecular-iris angle, iris-lens angle, and iris-lens contact distance (Chen et al., 2026) collected using a standardized topography device will all be included.
- A clinically significant rise in axial length over the follow up period or an increase in spherical equivalent refraction of at least 0.50 diopters annually will be considered myopia progression.
- Experienced optometrists would evaluate corneal topography images for traditional clinical analysis utilizing standard criteria such keratometric values, symmetry indices, corneal curvature patterns, and clinical judgment based on recognized guidelines.
- Based on these conventional evaluations, predictions about the course of myopia will be recorded.

AI-based analysis

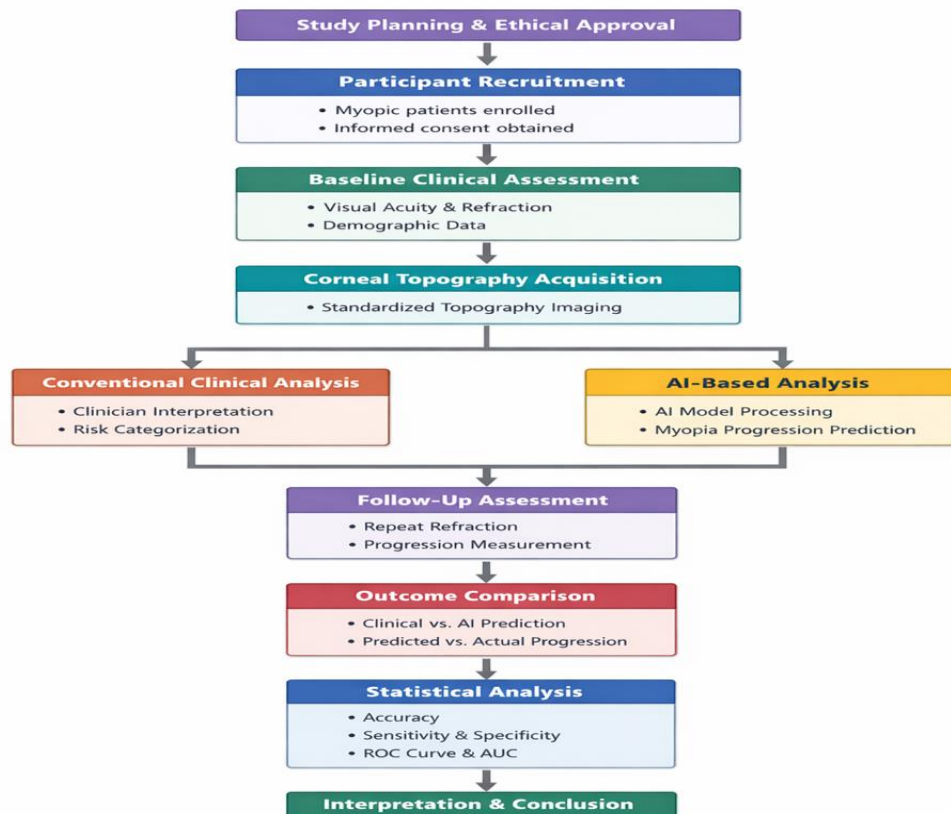
- The corneal topography parameters were analyzed using the Random Forest, Logistic Regression, and Decision Tree algorithms. Because Random Forest can reduce overfitting and is robust in handling multidimensional clinical datasets, it was chosen as the main model.
- Corneal topography data will be preprocessed for the machine learning method in order to eliminate noise and normalize input parameters. A machine learning will be fed pertinent characteristics such as corneal indices, elevation data, and curvature maps.
- The dataset will be separated into subsets for testing, validation, and training. Predictive outcome metrics like accuracy, sensitivity, specificity, and area under the receiver operating characteristic curve will be used to assess the model's performance. The actual clinical results seen during follow up will then be contrasted with the AI generated projections of myopia progression.

Comparison of machine learning and clinical analysis

The machine learning based system's prediction results will be compared to those obtained from conventional clinical trials. Appropriate statistical tests will be used to examine the degree of agreement between machine learning predictions and clinical evaluations. The capacity of each approach to precisely forecast myopia progression over time will serve as the main outcome measure. The consistency, objectivity, and repeatability of the predictions produced by both methods will be secondary outcomes.

Statistical analysis

Python based analytical tools or statistical software like SPSS will be used to analyze the data. The study population's clinical and demographic features will be compiled using descriptive statistics. Regression analysis and paired comparisons are two examples of inferential statistics that will be used to compare the predicted performance of clinical and AI based approaches. A statistically significant p-value is one that is less than 0.05.



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